

What can 1 week of seismic data
reveal about Strokkur's eruption
rhythm?

Seismic Analysis of the Strokkur Geyser

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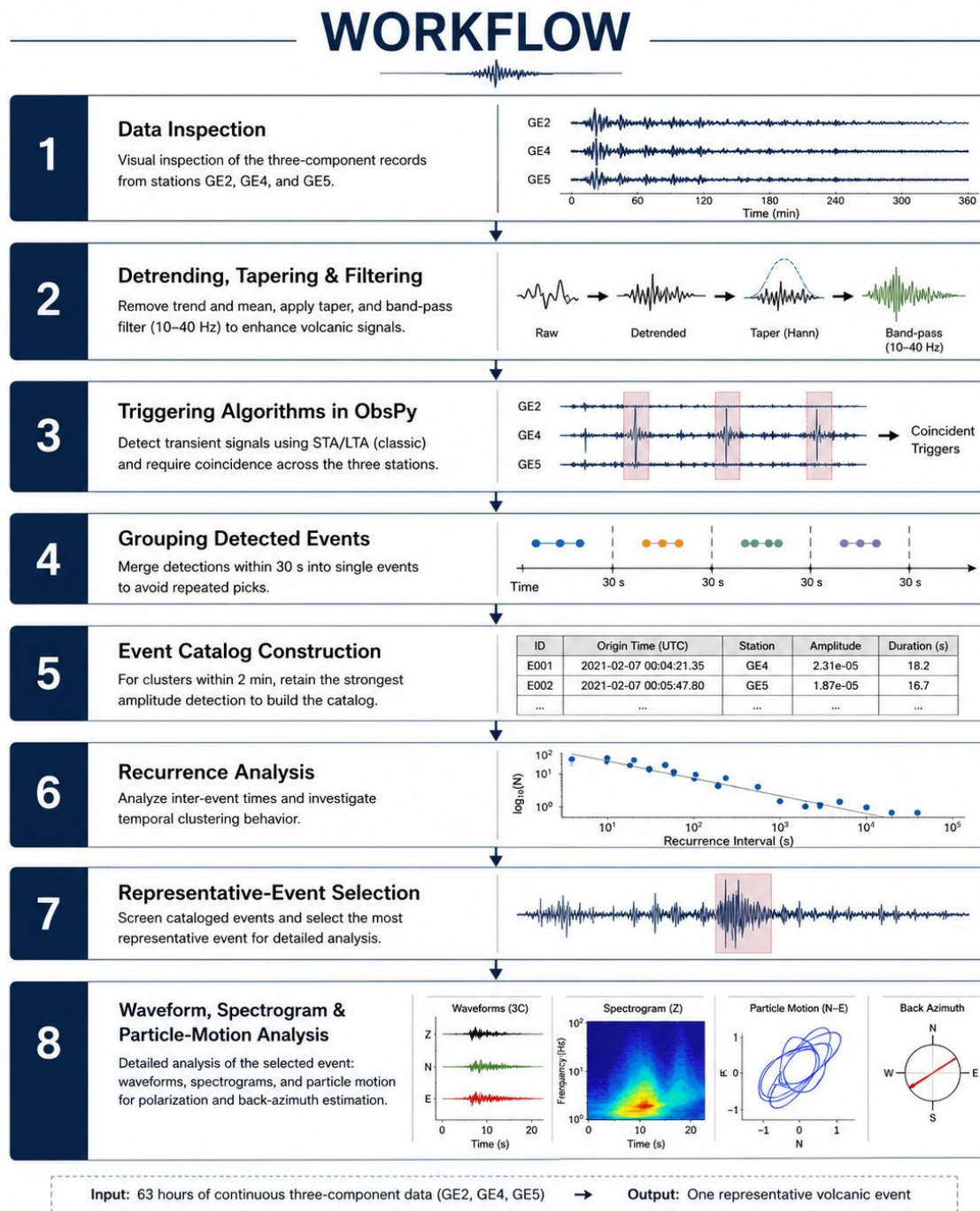
Seismological Data Science with Application to Volcano Systems



Strokkur Geyser

- **Location:** in the Haukadalur geothermal area, southwest Iceland.
- **Eruption:** repeatedly every few minutes, Heated groundwater, steam, and gas build pressure in the conduit.
- **Seismic production:** Bubble collapse and rapid fluid movement generate seismic vibrations.
- **Seismic Monitor usefulness:** enables continuous, nonvisual eruption detection and recurrence analysis.

Workflow



Initial Data Inspection

- Sampling rate: **100 Hz**.
- **7 days** of Seismic recordings.
- **63** Traces.

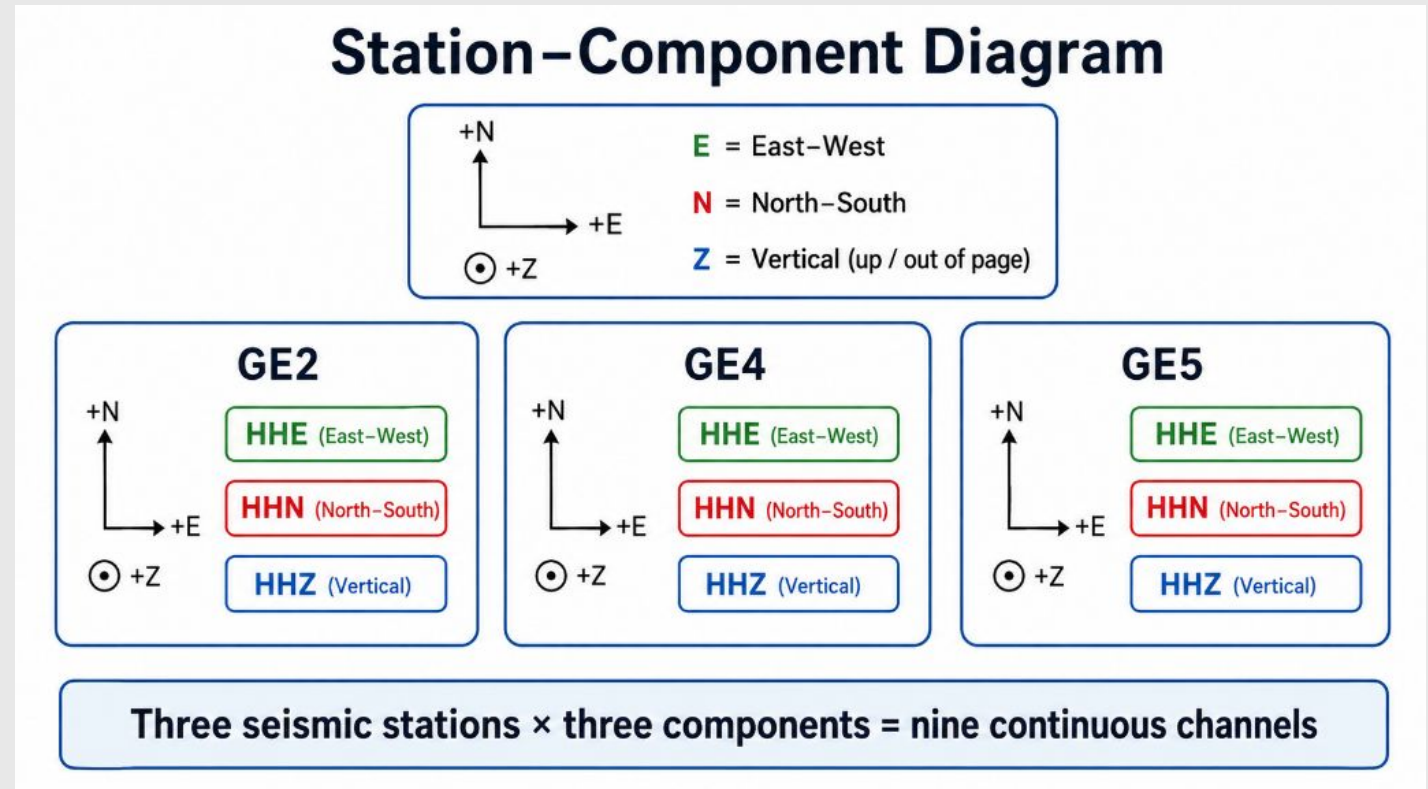


Figure created with generative AI for illustrative purposes.

preprocessing

- Demean
- Detrend
- Apply Hann Taper
- Apply 10-40 Hz band-pass Filter

-Preprocessing reduced drift, edge effects, and low-frequency noise.

Filter Selection

Why 10-40 Hz?

- Inspected **waveform, spectrum, & spectrogram.**
- Useful event energy mainly between **10-40 Hz.**
- Filter preserved the transient & **reduced background noise.**

Triggering Algorithms in ObsPy

- Applied to **continuous data**.
- Detected **sudden** seismic transients.
- Compared across **3** stations.
- **Retained** coherent network events.

Grouping detected events

```
[56]: catalog_week = pd.DataFrame(all_week_events)
```

```
catalog_week["time"] = pd.to_datetime(
    catalog_week["time"].astype(str)
)
```

```
catalog_week
```

```
[56]:
```

	day	time	duration	stations
0	1	2026-01-01 00:01:09.890000+00:00	3.74	[GE4, GE5, GE2]
1	1	2026-01-01 00:05:59.260000+00:00	3.55	[GE4, GE2, GE5]
2	1	2026-01-01 00:09:29.900000+00:00	5.07	[GE2, GE4, GE5]
3	1	2026-01-01 00:13:59.210000+00:00	3.58	[GE2, GE4, GE5]
4	1	2026-01-01 00:18:30.450000+00:00	3.67	[GE4, GE2, GE5]
...	...	...	...	...
3336	7	2026-01-07 10:59:54.910000+00:00	4.05	[GE5, GE2, GE4]
3337	7	2026-01-07 11:01:43.260000+00:00	6.10	[GE5, GE4, GE2]
3338	7	2026-01-07 11:05:25.070000+00:00	3.73	[GE4, GE2, GE5]
3339	7	2026-01-07 11:05:45.810000+00:00	3.24	[GE4, GE2, GE5]
3340	7	2026-01-07 11:05:55.470000+00:00	2.73	[GE4, GE2, GE5]

3341 rows × 4 columns

- **Closely spaced** detections combined.
- Grouping window: **30 S**.
- Compared across **3** stations.
- **Repeated** triggers treated as **1 episode**.
- **3341** transients—**2904** episodes.

Screenshot from the Jupyter Notebook analysis.

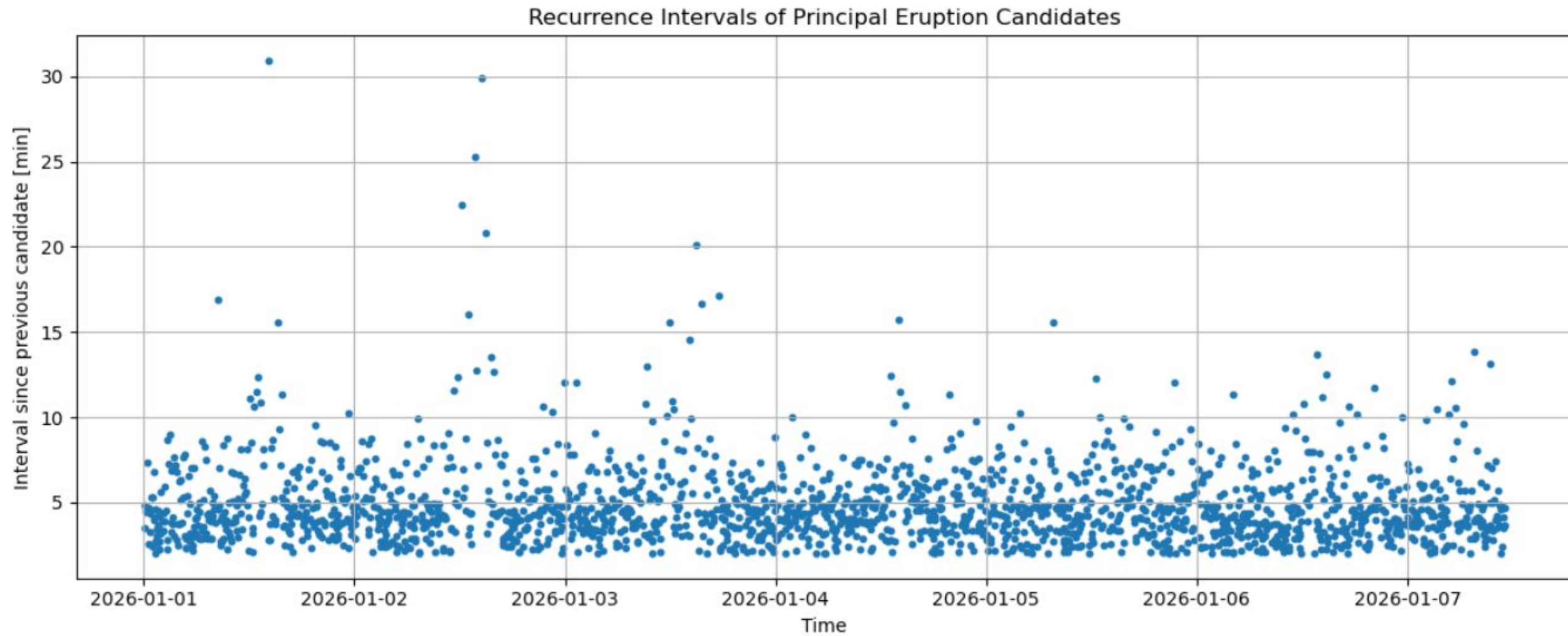
Event cataloging

- **Grouped** episodes within **2 m**.
- **Compared** network peak **amplitudes**.
- Kept the **strongest** event per cluster.
- **2904** episodes – **1946** principal candidates.

Recurrence analysis

- Stable rate: **12-13** candidates/hr.
- **Mean** Interval: **4.78** min.
- **Median** interval: **4.27** min.
- **Most** interval: **3-6** min.

-Recurrence seismic activity remained relatively regular throughout the week

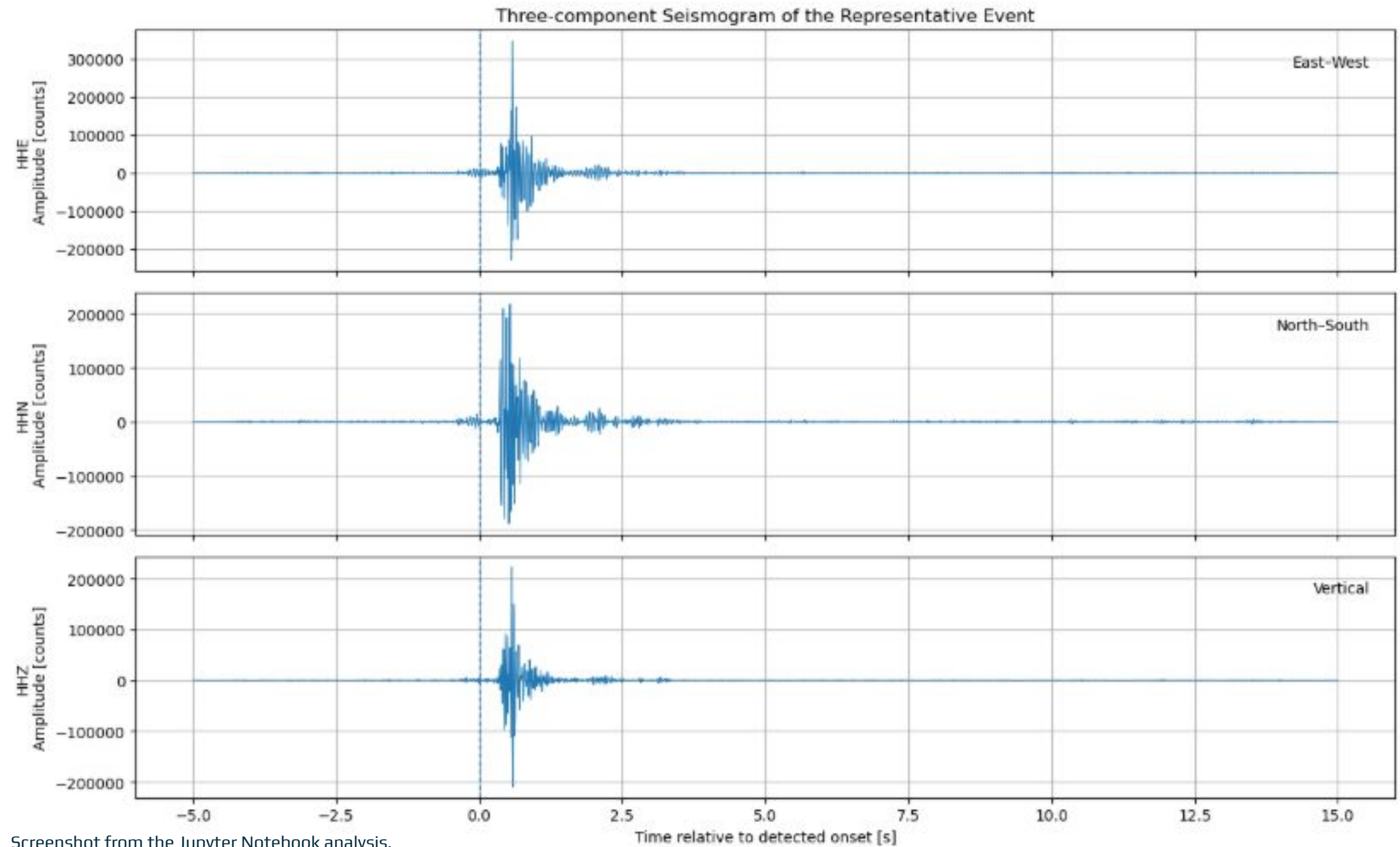


Screenshot from the Jupyter Notebook analysis.

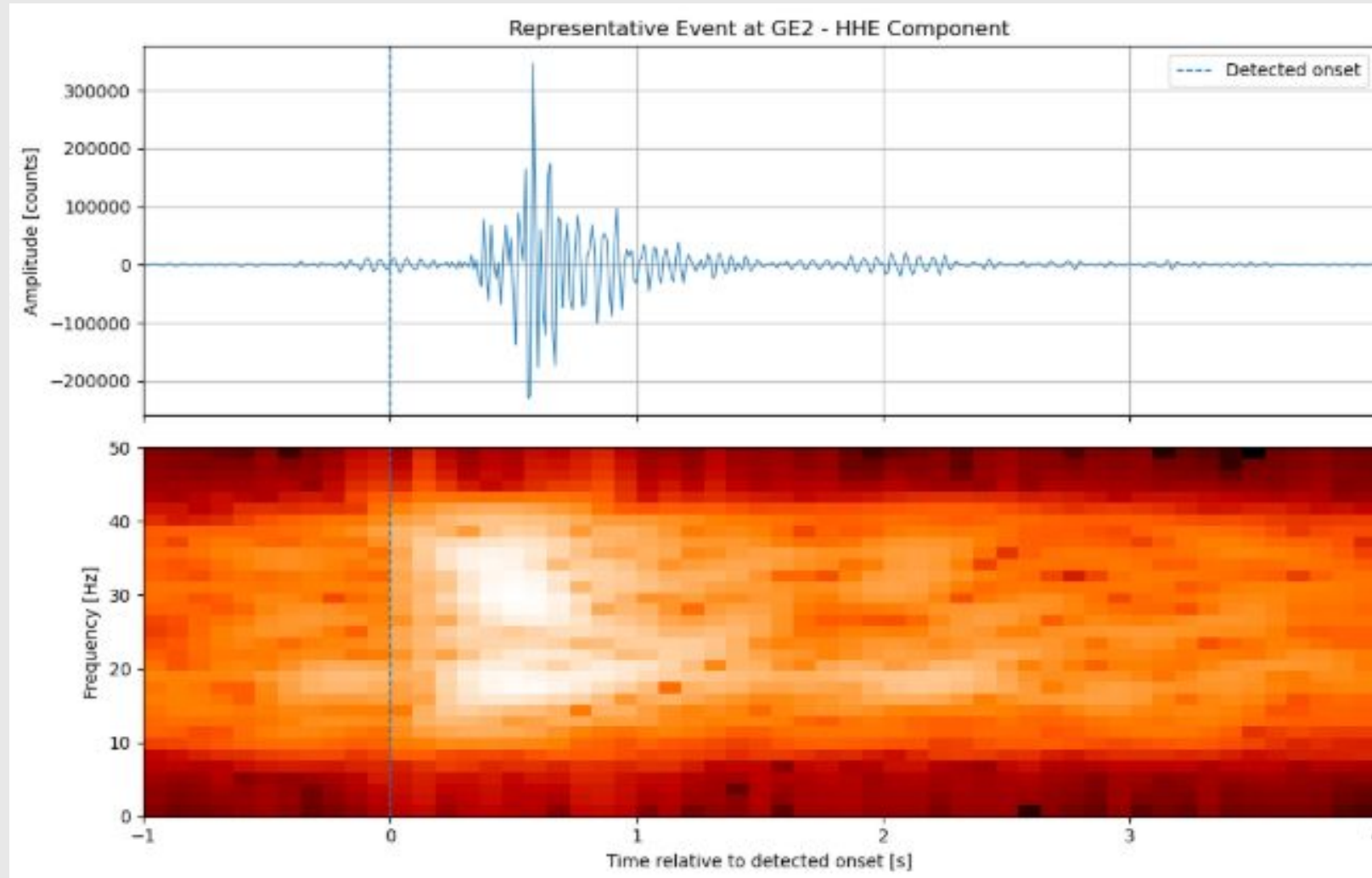
Rep., event selection

- Selected from **final** candidate **catalog**.
- Clear at all **3** stations.
- **Strong** signal to noise **ratio**.
- **Visible** on HHE, HHN, HHZ.

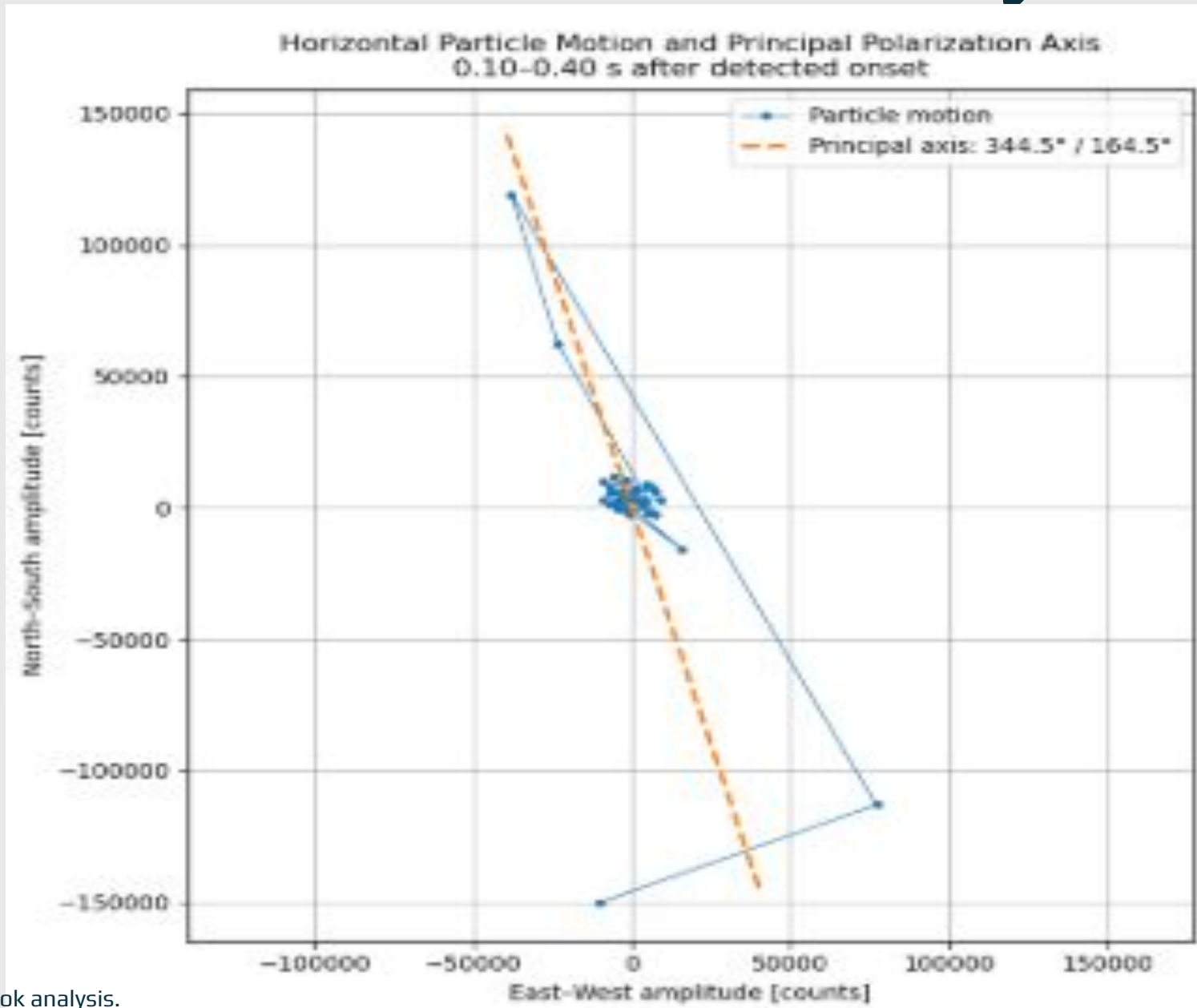
-Selected event: 2 January 2026 – 09:50:23.67 UTC



Waveform & Spectrogram



Particle motion analysis



Interpretation

The data show regular, persistent seismic activity, with most candidates repeating every 3–6 minutes and the selected event showing a clear waveform, frequency signal, and directional motion.

Limitations

- Candidates were not **visually** confirmed eruptions.
- **No** station **vent coordinates** were provided.
- **Back azimuth** retained a **180°** ambiguity.
- The **30 s** and **2 min** thresholds were project-specific choices.
- **Day 7** contained only about **12 hours** of data.

Conclusions

- **Continuous** seismic monitoring successfully detected recurring Strokkur-related events **throughout the week**. The **principal candidates** occurred mainly every **3–6 minutes**, while **waveform, spectrogram, and particle-motion** analysis clearly characterized **one** representative **event**. The results show that seismic data can effectively monitor regular geyser activity, although independent **visual confirmation** and station **coordinates** would **improve the interpretation**.

References

- Eibl, E. P. S., et al. (2020). *Eruption interval monitoring at Strokkur Geyser, Iceland*. **Geophysical Research Letters**, 47, e2019GL085266.
- Eibl, E. P. S., et al. (2024). *Video camera and seismic monitoring of water-bulge explosions at Strokkur Geyser, Iceland*. **Volcanica**, 7(1).
- GFZ German Research Centre for Geosciences. (2020). *Seismological experiment at Strokkur from 2020—Network 2Z*.
- ObsPy Development Team. *ObsPy documentation: Python framework for processing seismological data*.

Thank you!

Do not be hesitant to ask any **Questions!**